

MAAT String Selection II

Structural Path Selection in a Reduced KKLT Landscape

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Abstract

The first MAAT string-selection benchmark treated candidate string backgrounds as endpoints and ranked them by structural support. In particular, reduced IIB/KKLT bridge tests showed that energy-only ranking and structural ranking need not select the same vacua: tadpole closure, metastability, and EFT-control diagnostics can reorder the preferred subset.

This paper extends the framework from static vacuum ranking to structural path selection. Candidate vacua are represented as nodes B_i in a reduced landscape graph, and possible transitions $B_i \rightarrow B_j$ are represented as directed edges between near neighbours in flux/moduli feature space. Each edge is assigned a phenomenological path action

$$\mathcal{A}_{ij} = B_{ij} + \lambda_F \max(F_{\text{MAAT}}[B_j] - F_{\text{MAAT}}[B_i], 0) + \lambda_c C_{ij} + \lambda_Q Q_{ij} + \lambda_R R_{ij}^{\text{loss}},$$

where B_{ij} is a barrier proxy, C_{ij} an EFT-control penalty, Q_{ij} a tadpole/constraint penalty, and R_{ij}^{loss} a stability-loss penalty. The corresponding transition weight is

$$P(B_i \rightarrow B_j) \propto \Gamma_{ij} \exp[-\mathcal{A}_{ij}], \quad \Gamma_{ij} = \exp[-B_{ij}].$$

Using the existing reduced IIB/KKLT bridge ensemble, we build a graph of 77 minima and 462 directed near-neighbour transitions. The static energy-best minimum is not the structural-best minimum. The graph identifies a two-edge structural path from the energy-best node to the structural-best node,

$$60 \rightarrow 22 \rightarrow 30,$$

with structural path cost 0.9079. The top 20 structural transition edges have zero overlap with the top 20 energy-action edges, and 53.9% of all graph edges are downhill in F_{bridge} .

The result is not a Coleman–De Luccia decay calculation and not a microscopic string tunnelling rate. Its contribution is narrower: MAAT string selection is extended from vacuum ranking to a reproducible structural landscape-flow diagnostic.

1 Purpose and Status

The previous string-landscape paper introduced MAAT as a structural ranking layer over candidate string backgrounds. That construction asks:

Which candidate vacua are structurally preferred?

The present paper asks the next question:

Which transitions between candidate vacua are structurally preferred?

This is a phenomenological benchmark. It does not compute Coleman–De Luccia instantons, brane nucleation rates, full ten-dimensional backreaction, or microscopic string decay amplitudes. Instead, it defines a reduced transition graph over already-generated KKLT bridge candidates and asks whether the resulting structural flow differs from energy-only flow.

2 From Ranking to Paths

In the static ranking problem, each candidate background B_i is assigned a structural functional

$$F_{\text{MAAT}}[B_i] = -\sum_a \lambda_a \log(\epsilon + \Gamma_a[B_i]), \quad \Gamma_a[B_i] = \frac{1}{1 + d_a[B_i]}. \quad (1)$$

Lower F_{MAAT} indicates larger structural support.

For paths, the relevant object is not only $F_{\text{MAAT}}[B_i]$, but the transition from one background to another. Define

$$\Delta F_{\text{MAAT}}^{ij} = F_{\text{MAAT}}[B_j] - F_{\text{MAAT}}[B_i]. \quad (2)$$

An uphill transition in F_{MAAT} is structurally suppressed, while a downhill transition is favoured but still limited by a transition barrier.

The schematic transition law is

$$P(B_i \rightarrow B_j) \propto \Gamma_{ij} \exp[-\lambda_F \max(\Delta F_{\text{MAAT}}^{ij}, 0)]. \quad (3)$$

The present benchmark refines this by adding control, tadpole, and stability terms.

3 Reduced KKLT Path Graph

The source data are the existing IIB/KKLT bridge candidates from the string-landscape benchmark. Only candidates classified as minima are used as nodes:

$$B_i \in \{\text{AdS minimum, dS minimum}\}.$$

Unstable stationary points are excluded from the path graph.

Each node carries:

- integer flux vectors (F, H) ,
- D3-tadpole residual ΔQ_{D3} ,
- reduced KKLT parameters (W_0, D, τ) ,
- vacuum energy V_i ,
- metastability data through $\lambda_{\min}$,
- structural scores F_{bridge} and F_{pheno} .

Edges are constructed by nearest neighbours in a standardized feature space containing flux coordinates, τ , W_0 , uplift D , energy, and tadpole residual. For each node, the $k = 6$ nearest neighbours define directed candidate transitions.

4 Path Action

The path action used in the benchmark is

$$\mathcal{A}_{ij} = B_{ij} + \lambda_F \max(F_{\text{bridge}}[B_j] - F_{\text{bridge}}[B_i], 0) + \lambda_c C_{ij} + \lambda_Q Q_{ij} + \lambda_R R_{ij}^{\text{loss}}. \quad (4)$$

The barrier proxy is

$$B_{ij} = w_f d_f^{ij} + w_\tau d_\tau^{ij} + w_W d_W^{ij} + w_E d_E^{ij} + w_Q d_Q^{ij}. \quad (5)$$

The transition probability is normalized over all outgoing edges from a source node:

$$P_{ij}^{\text{struct}} = \frac{\exp[-\mathcal{A}_{ij}]}{\sum_{k \in \mathcal{N}(i)} \exp[-\mathcal{A}_{ik}]}. \quad (6)$$

For comparison, an energy-only action is also computed:

$$\mathcal{A}_{ij}^E = B_{ij} + 0.85 \frac{\max(|V_j| - |V_i|, 0)}{\max(|V_i|, |V_j|, \epsilon)}. \quad (7)$$

Table 1: Path-action parameters used in the benchmark.

Parameter	Value
k nearest neighbours	6
λ_F	0.85
λ_c	1.10
λ_Q	0.65
λ_R	0.75
$(w_f, w_\tau, w_W, w_E, w_Q)$	(0.45, 0.20, 0.15, 0.10, 0.10)

5 Numerical Results

Table 2: Reduced KKLt path-selection graph.

Quantity	Result
Minima nodes	77
Directed near-neighbour edges	462
AdS minima	45
dS minima	32
Spearman $ V $ vs. F_{bridge} among minima	0.1271
Spearman ΔQ_{10} vs. F_{bridge} among minima	0.9486
Fraction of structurally downhill edges	0.539
Top-20 structural/energy edge overlap	0/20

The low Spearman correlation between $|V|$ and F_{bridge} confirms that the structural ranking is not simply an energy ranking. The high correlation between tadpole residual and F_{bridge} is expected: the bridge functional explicitly rewards tadpole closure and EFT control.

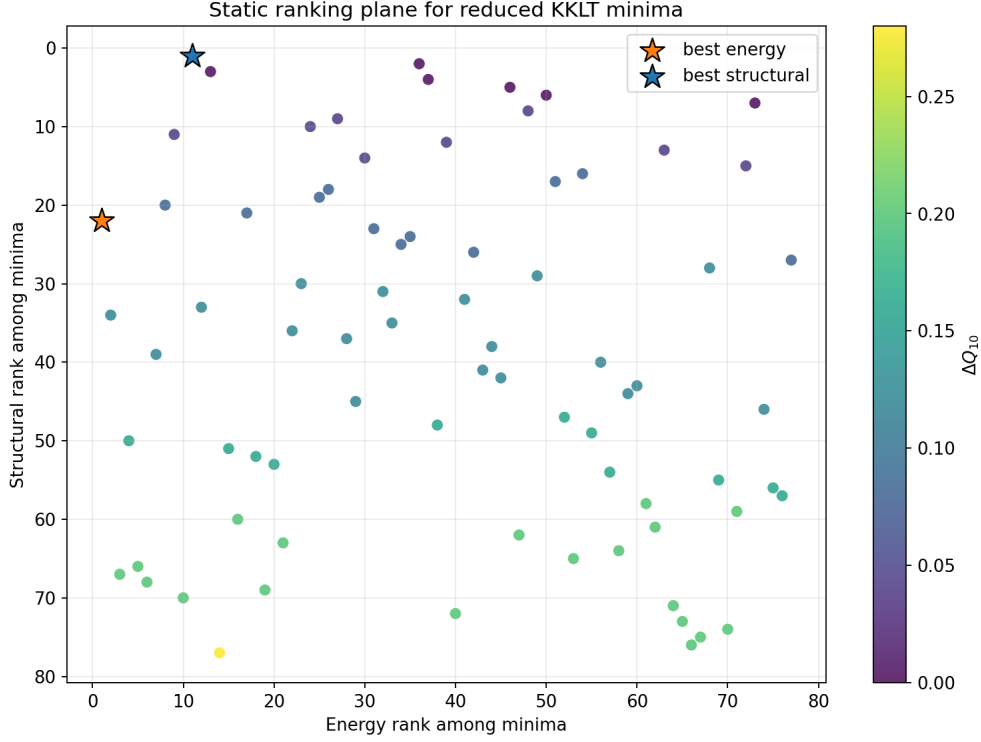


Figure 1: Static ranking plane for reduced KKL minima. The energy-best minimum and structural-best minimum are distinct.

6 Structural Path from Energy-Best to Structural-Best

The energy-best minimum has candidate ID 128 and node ID 60. The structural-best minimum has candidate ID 67 and node ID 30. The structural shortest path between them is

$$\boxed{60 \rightarrow 22 \rightarrow 30.} \quad (8)$$

Its total structural path cost is

$$\mathcal{A}_{60 \rightarrow 22 \rightarrow 30} = 0.9079.$$

The same target path evaluated using the energy-action cost has cost 1.7191.

Table 3: Selected structural path from energy-best to structural-best node.

Edge	$F_{\text{bridge}}(i)$	$F_{\text{bridge}}(j)$	ΔF_{bridge}	$\mathcal{A}_{ij}$
$60 \rightarrow 22$	-1.9947	-1.9957	-0.0010	0.4769
$22 \rightarrow 30$	-1.9957	-4.3327	-2.3371	0.4310

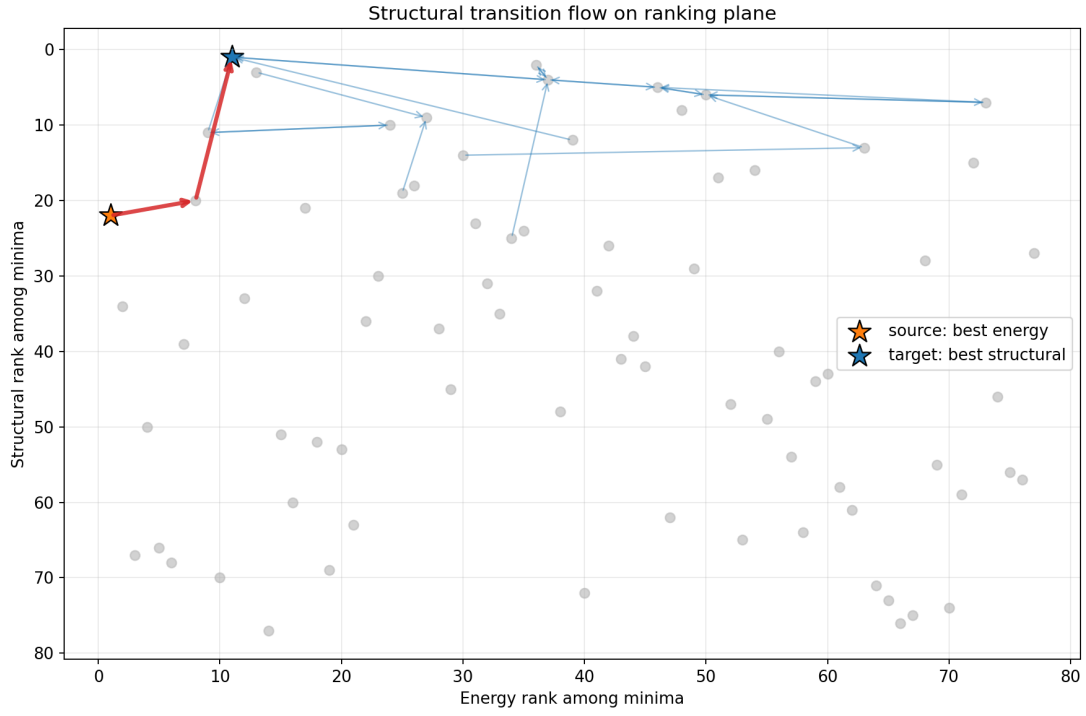


Figure 2: Structural transition flow on the ranking plane. Blue arrows show high-probability structural transitions; red arrows mark the selected structural path from the energy-best node to the structural-best node.

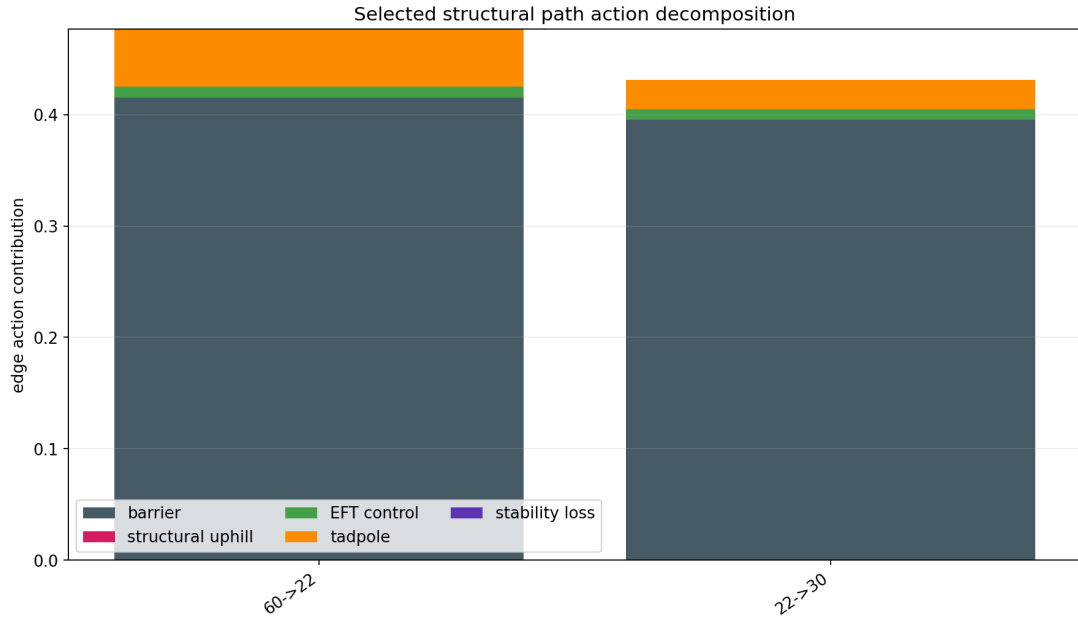


Figure 3: Action decomposition for the selected path. The structural path is not controlled by energy alone; barrier, control, and tadpole terms contribute to the transition cost.

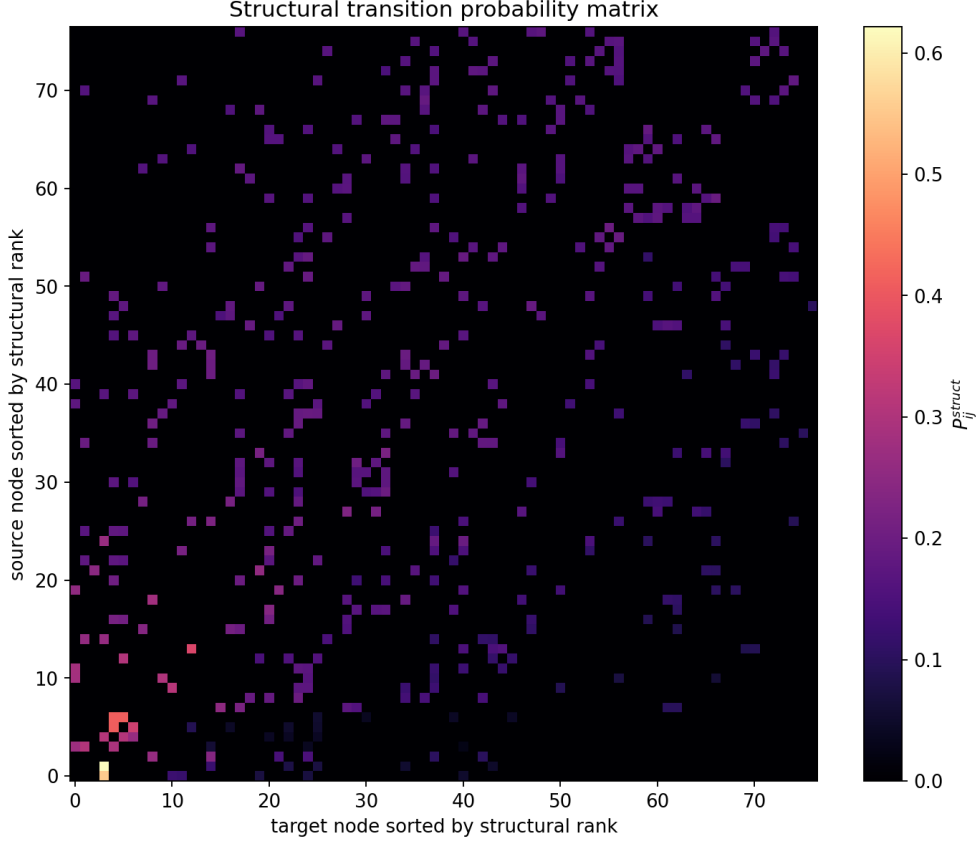


Figure 4: Structural transition probability matrix sorted by structural rank. The sparse matrix encodes a reduced landscape-flow diagnostic rather than a thermal equilibrium distribution.

7 Energy Paths versus Structural Paths

The top 20 structural transition edges have no overlap with the top 20 energy-action edges. This is the central diagnostic result:

$$\text{Top20}(P_{ij}^{\text{struct}}) \cap \text{Top20}(P_{ij}^E) = \emptyset. \quad (9)$$

Thus, even on the same node set and the same near-neighbour graph, structural path selection and energy-only path selection identify different preferred transitions.

Greedy paths from the top energy-ranked sources show the same separation. For example, the energy-best node follows the structural path

$$60 \rightarrow 22 \rightarrow 30 \rightarrow 14 \rightarrow 29 \rightarrow 14,$$

whereas the energy-action path follows

$$60 \rightarrow 74 \rightarrow 27 \rightarrow 74.$$

The structural path moves toward low F_{bridge} nodes, while the energy-action path remains trapped in an energy-local cycle.

8 Interpretation

The benchmark supports the following limited claim:

MAAT extends string vacuum ranking into structural landscape flow: preferred endpoints and preferred transition paths need not be determined by energy alone.

This is not yet a theory of vacuum decay. However, it is a useful intermediate object. It allows one to ask whether a landscape contains structurally attractive basins, whether energy-local minima trap paths away from structurally better regions, and whether tadpole/control-improving transitions form coherent flows.

9 Limitations

- The transition graph is a reduced proxy graph, not a full string landscape.
- The barrier $\Gamma_{ij} = \exp[-B_{ij}]$ is a phenomenological proxy, not a computed instanton action.
- No Coleman–De Luccia, brane nucleation, or flux tunnelling amplitude is computed.
- Full 10D backreaction, warping, local sources, orientifolds, and Bianchi identities are not solved.
- The graph uses the reduced IIB/KKLT bridge dataset; exact-period, multi-modulus, LVS, and swampland extensions remain future work.

10 Reproducibility

The complete experiment is included in the Structural Selection Principle repository:

Repository URL:

https://github.com/Chris4081/structural-selection-principle/tree/main/experiments/string_landscape_path_selection

Run from the experiment folder:

```
python3 maat_string_path_selection_kklt_v01.py
```

The script writes:

- `outputs_path_selection/path_selection_summary.json`
- `outputs_path_selection/path_selection_nodes.csv`
- `outputs_path_selection/path_selection_edges.csv`
- `outputs_path_selection/path_selection_greedy_paths.csv`
- `outputs_path_selection/best_energy_to_best_bridge_path.csv`
- `outputs_path_selection/fig1_static_ranking_plane.png`
- `outputs_path_selection/fig2_structural_flow_graph.png`
- `outputs_path_selection/fig3_path_action_decomposition.png`
- `outputs_path_selection/fig4_transition_probability_matrix.png`

Data attribution and license note. The experiment uses derived synthetic KKLT-bridge candidates generated by the repository’s string-landscape benchmark scripts. No external survey data or proprietary dataset is redistributed. The benchmark is a reproducibility artifact for the MAAT structural-selection framework and is not endorsed by the authors of the original string-theory literature cited below.

11 Conclusion

MAAT String Selection II extends structural selection from static vacuum ranking to structural path selection. In a reduced KKLT bridge landscape, the energy-best node and structural-best node are distinct, but the graph contains a two-edge structural path between them. Top structural transition edges differ completely from top energy-action edges, demonstrating that structural path selection is not merely energy ranking in disguise.

The next natural step is to repeat the construction on exact-period and multi-modulus ensembles and eventually replace the barrier proxy by a microscopic tunnelling calculation.

References

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